

ESTIMATING THE SHAPE PARAMETER OF THE WEIBULL DISTRIBUTION BY SHRINKAGE TOWARDS AN INTERVAL

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Summary: This paper proposes some shrunken estimators for the shape parameter of the Weibull distribution under censored sampling when some *a priori* or guessed interval containing the parameter β is available. Comparisons with the usual unbiased estimator, in terms of mean squared error, is made. It is found that the proposed estimators are preferable to the usual estimator in some guessed interval of the parameter space of β .

1. INTRODUCTION

In failure situations where the system is composed of numerous components in which a large number of flaws exist and the failure depends on its severest flaw (or weakest spot), the Weibull model of failure time is found to be most appropriate. The p.d.f. of a failure time X following a two-parameter Weibull distribution is given by

$$(1) \quad f(x, \theta, \beta) = \left(\frac{\beta}{\theta}\right) \left(\frac{x}{\theta}\right)^{\beta-1} \exp\left\{-\left(\frac{x}{\theta}\right)^{\beta}\right\}, \quad (x > 0, \theta, \beta > 0).$$

The parameters θ and β are called the characteristic life and the shape parameter respectively. The variable $Y = \log_e X$

follows an extreme value distribution with cumulative distribution function

$$(2) \quad G(y) = 1 - \exp\left\{-\exp\left(\frac{y-u}{b}\right)\right\}, \quad (-\infty < y < \infty, -\infty < u < \infty, b > 0)$$

where $b = \frac{1}{\beta}$ and $u = \log_e \theta$ are respectively the scale and location parameters of the above extreme value distribution. Suppose $x_1, x_2, \dots, x_r$ denote the r smallest ordered observations in a sample of size n from the Weibull distribution and $y_1, y_2, \dots, y_r$ where $y_i = \log_e x_i$ represent the r smallest observations in a sample of size n from the extreme value distribution. Bain (1972) proposed the following unbiased estimator for b as

$$\hat{b} = -\frac{r-1}{\sum_{i=1}^{r-1} (y_i - y_r)/n K_{r,n}},$$

where $K_{r,n} = -\frac{1}{n} E \sum_{i=1}^{r-1} (w_i - w_r)$ and $w_i = \frac{y_i - u}{b}$ are ordered variables from an extreme value distribution with $u = 0$ and $b = 1$. The estimator $\hat{b}$ was found to have high relative efficiency for heavily censored cases. It was also shown that the asymptotic relative efficiency of $\hat{b}$ is zero for the complete sample case. Engelhardt and Bain (1973) proposed a general form of the estimator as

$$\hat{b}_s = \frac{r}{\sum_{i=1}^r |y_i - y_r|/n K_{s,r,n}},$$

where s is chosen to minimize the variance of $\hat{b}_s$ and $K_{s,r,n}$ is an unbiased constant. Both small sample and large sample properties of $\hat{b}_s$ were studied and it was found that for $\frac{r}{n} \leq 0.9$ or $n \leq 15$, $\hat{b}_s$ reduces to $\hat{b}$.

A chi-square approximation for the distribution of $\frac{\hat{b}}{b}$ is given in Bain (1972) by fitting one moment. This approximation was found to be good for heavily censored samples. In Engelhardt,

and Bain (1973), a chi-square approximation was determined by fitting two moments and this approximation was shown to be quite good for all censoring levels and any sample size. More specifically it was found that $\left[\frac{hb_s}{b}\right] \sim \chi^2(h)$ where $h = 2/\text{var}(\hat{b}_s/b)$. The expressions for the determination of $h = 2/\text{var}(\hat{b}_s/b)$ are given in Engelhardt and Bain (1973), Mann and Fertig (1975) and Engelhardt (1975). The above chi-square approximation has the correct first two moments for small n and it approaches the proper asymptotic normal distribution. In Engelhardt and Bain (1973), the accuracy of the approximation was also investigated for small n by the use of Monte Carlo methods. It was found that the amount of discrepancy between the approximate and true percentages appeared to be no more than the amount of fluctuations in the Monte Carlo values themselves. Engelhardt (1975) pointed out that his expressions lead to more accurate values of h .

Due to considerable handling of manufactured items in the past one may have a guessed value of the shape parameter β in the form of either a point or an interval. Thompson (1968a), (1968b) introduced the method of shrinking the usual estimator towards the guessed point value or an interval guessed value. Considering the knowledge of a point guessed value of β , Singh and Bhatkulikar (1978) have suggested two preliminary test estimators for the shape parameter β .

In this paper we consider the problem of estimating the shape parameter β of the Weibull distribution using 'shrinkage technique' in case *a priori* information about β is available in the form that β lies between β_1 and β_2 (β_1 and β_2 are known values). Some shrunken estimators are proposed and compared with the usual unbiased estimator in terms of mean squared error.

2. PROPOSED ESTIMATORS AND THEIR PROPERTIES

Suppose n items are put to life test until r failures are observed at times $x_1, x_2, \dots, x_r$ which follow the Weibull distribution given in (1). The usual unbiased estimator of β based on these observations is

$$\hat{\beta} = \frac{h-2}{t} \quad \text{which has} \quad \text{var}(\hat{\beta}) = \frac{2\beta^2}{(h-4)}.$$

Following Thompson (1968a) and (1968b), one can define an estimator

$$(3) \quad M_L = K\left(\frac{h-2}{t}\right) + (1-K)\left(\frac{\beta_1 + \beta_2}{2}\right)$$

where K is a constant lying between zero and one and is to be specified by the experimenter according to his belief in $\frac{\beta_1 + \beta_2}{2}$. The value of K near zero implies strong belief in

$\frac{\beta_1 + \beta_2}{2}$. Now, since $\left\{\frac{h\hat{\beta}_s}{b}\right\}$ follows approximately a chi-

square distribution with h degrees of freedom, we have

$$(4) \quad \text{MSE}(M_L) \approx \frac{2\beta^2}{(h-4)} \left[K^2 + \frac{(h-4)(1-K)^2}{2} \left\{ 1 - \frac{\beta_1 + \beta_2}{2\beta} \right\}^2 \right].$$

The relative efficiency of the estimator M_L with respect to the unbiased estimator $\hat{\beta}$ is defined as

$$(5) \quad \text{REF}(M_L, \hat{\beta}) = \frac{\text{MSE}(\hat{\beta})}{\text{MSE}(M_L)} = \left[K^2 + \frac{(h-4)(1-K)^2}{2} \left\{ 1 - \frac{\beta_1 + \beta_2}{2\beta} \right\}^2 \right]^{-1}.$$

The relative efficiency of the estimator M_L has been calculated for $n = 20$, $r = 4(4)12$, $K = 0.2(0.2)0.8$ and for different values of $\frac{\beta_1}{\beta}$ and $\frac{\beta_2}{\beta}$ and are presented in table 1. From table 1, we observe that the relative efficiency reaches the

maximum when K is small and $\frac{\beta_1 + \beta_2}{2\beta} \approx 1$, it decreases as the difference between β and $\frac{\beta_1 + \beta_2}{2}$ increases. If we have very little confidence in our guessed interval (β_1, β_2) , the proposed estimator M_L is not preferable. It is easy to conclude from the table that the proposed estimator M_L is preferable to $\hat{\beta}$ if

- (i) $0 \leq K \leq 0.80$
- (ii) $1.00 \leq \frac{\beta_1 + \beta_2}{2\beta} \leq 1.25$ and
- (iii) censoring fraction r/n is small.

Following Pandey (1979), we can propose a modified ratio type estimator for the shape parameter β of the Weibull distribution as follows:

$$(6) \quad M_w = \hat{\beta} \left[\frac{W\beta_1 + (1-W)\beta_2}{\hat{\beta}} \right]^{(1-K)}, \quad 0 \leq W \leq 1,$$

where W is a constant. Now,

$$(7) \quad \begin{aligned} \text{MSE}(M_w) \approx \beta^2 & \left[\left(\frac{h-2}{2} \right)^K (\Gamma(h/2))^{-1} \left(\frac{W\beta_1 + (1-W)\beta_2}{\beta} \right)^{1-K} \right. \\ & \left. \left\{ \left(\frac{h-2}{2} \right)^K \left(\frac{W\beta_1 + (1-W)\beta_2}{\beta} \right)^{1-K} \Gamma\left(\frac{h}{2} - 2K\right) - 2\Gamma\left(\frac{h}{2} - K\right) \right\} + 1 \right]. \end{aligned}$$

The relative efficiency of the estimator M_w with respect to the unbiased estimator $\hat{\beta}$ is

$$(8) \quad \text{REF}(M_w, \hat{\beta}) = \frac{\text{MSE}(\hat{\beta})}{\text{MSE}(M_w)}.$$

The relative efficiency of the estimator M_w has been given in table 2 for $n = 20$, $r = 4(4)12$, $K = 0.2(0.2)0.8$, $W = 0.2(0.6)0.8$

and for different values of $\frac{\beta_1}{\beta}$ and $\frac{\beta_2}{\beta}$. From this table, we observe that the estimator M_w has smaller mean squared error than the unbiased estimator $\hat{\beta}$ if

- (i) $0 \leq k \leq 0.8$
- (ii) $0.60 \leq \frac{\beta_1 + \beta_2}{2\beta} \leq 1.00$
- (iii) W is small, and
- (iv) censoring fraction r/n is small.

It is also noticed that the estimator M_w may have smaller mean squared error than the estimator $\hat{\beta}$ even if $\frac{\beta_1 + \beta_2}{2\beta} > 1$ when W is taken near to one.

We can also propose a new type of shrunken estimator for the shape parameter β of the Weibull distribution as

$$(9) \quad M_d = \begin{cases} \beta_1 & \text{if } t > \frac{h-2}{\beta_1} \\ k\left(\frac{h-2}{t}\right) + (1-k)\frac{\beta_1 + \beta_2}{2} & \text{if } \frac{h-2}{\beta_2} \leq t \leq \frac{h-2}{\beta_1} \\ \beta_2 & \text{if } t < \frac{h-2}{\beta_2} \end{cases}$$

which has

$$(10) \quad \begin{aligned} \text{MSE}(M_d) \approx & \beta^2 \left[\left(\frac{\beta_1}{\beta} - 1 \right)^2 - \frac{\beta_1}{\beta} \left(\frac{\beta_1}{\beta} - 2 \right) I(x_0, \frac{h}{2}) \right. \\ & + \frac{\beta_2}{\beta} \left(\frac{\beta_2}{\beta} - 2 \right) I(x_1, \frac{h}{2}) + \frac{K^2(h-2)}{(h-4)} \left\{ I(x_0, \frac{h}{2} - 2) - I(x_1, \frac{h}{2} - 2) \right\} \\ & + \left(\frac{K(1-K)(\beta_1 + \beta_2)}{\beta} - 2K \right) \left\{ I(x_0, \frac{h}{2} - 1) - I(x_1, \frac{h}{2} - 1) \right\} \\ & \left. + \frac{(1-K)(\beta_1 + \beta_2)}{\beta} \left(\frac{(1-K)(\beta_1 + \beta_2)}{4\beta} - 1 \right) \left\{ I(x_0, \frac{h}{2}) - I(x_1, \frac{h}{2}) \right\} \right] \end{aligned}$$

where $x_0 = \frac{(h-2)\beta}{2\beta_1}$, $x_1 = \frac{(h-2)\beta}{2\beta_2}$ and $I(x, p) = \frac{1}{\Gamma(p)} \int_0^x e^{-u} u^{p-1} du$.

The relative efficiency of the estimator M_d with respect to $\hat{\beta}$ is defined as

$$(11) \quad \text{REF}(M_d, \hat{\beta}) = \frac{\text{MSE}(\hat{\beta})}{\text{MSE}(M_d)}.$$

The relative efficiency of the estimator M_d has been obtained for $n=20$, $r=4(4)12$, $K=0.2(0.2)0.8$ and for different values of $\frac{\beta_1}{\beta}$ and $\frac{\beta_2}{\beta}$ and are summarized in table 1. We observe from this table that the estimator M_d has smaller mean squared error than $\hat{\beta}$ if

$$(i) \quad 0 \leq K \leq 0.80$$

$$(ii) \quad 0.60 \leq \frac{\beta_1 + \beta_2}{2\beta} \leq 1.50 \quad \text{and}$$

(iii), censoring fraction r/n is small.

In particular, if $K=1$, the estimator M_d reduces to M'_d where

$$(12) \quad M'_d = \begin{cases} \beta_1 & \text{if } t > \frac{h-2}{1} \\ \frac{h-2}{t} & \text{if } \frac{h-2}{\beta_2} \leq t \leq \frac{h-2}{\beta_1} \\ \beta_2 & \text{if } t < \frac{h-2}{\beta_2} \end{cases}$$

which has

$$(13) \quad \text{MSE}(M'_d) \approx \beta^2 \left[\left(\frac{\beta_1}{\beta} - 1 \right)^2 - \frac{\beta_1}{\beta} \left(\frac{\beta_1}{\beta} - 2 \right) I(x_0, \frac{h}{2}) + \frac{\beta_2}{\beta} \left(\frac{\beta_2}{\beta} - 2 \right) I(x_1, \frac{h}{2}) + \frac{(h-2)}{(h-4)} \{ I(x_0, \frac{h}{2} - 2) - I(x_1, \frac{h}{2} - 2) \} - 2 \{ I(x_0, \frac{h}{2} - 1) - I(x_1, \frac{h}{2} - 1) \} \right].$$

The relative efficiency of the estimator M'_d with respect to $\hat{\beta}$ is defined as

$$(14) \quad \text{REF}(M'_d, \hat{\beta}) = \frac{\text{MSE}(\hat{\beta})}{\text{MSE}(M'_d)}.$$

The relative efficiency of the estimator M'_d has been given in table 1 for $n = 20$, $r = 4(4)12$ and for different values of $\frac{\beta_1}{\beta}$ and $\frac{\beta_2}{\beta}$. From the table, we observe that the estimator M'_d has smaller mean squared error than $\hat{\beta}$ if

$$(i) \quad 0.60 \leq \frac{\beta_1 + \beta_2}{2\beta} \leq 3.00 \quad \text{and}$$

(ii) censoring fraction r/n is small.

Comparing the relative efficiencies of the estimators M_d , M_L , M_w and M'_d given in tables 1 and 2, we observe that

if (i) $0 \leq K \leq 0.80$ (ii) $0.60 \leq \frac{\beta_1 + \beta_2}{2\beta} \leq 1.50$ and (iii) censoring fraction r/n is small, the estimator M_d has greater efficiency than M_L and M_w . Therefore the estimator M_d is preferable. But if K is near to one and r/n is small M'_d is preferable.

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TABLE 1. Relative efficiencies of M_L , M_d and M'_d with respect to the usual unbiased estimator $\hat{\beta}$.

r	$\frac{\beta_1}{\beta}$	$\frac{\beta_2}{\beta}$	K = 0,20		K = 0,40		K = 0,60		K = 0,80		M'_d
			M_L	M_d	M_L	M_d	M_L	M_d	M_L	M_d	
4	0,20	0,40	2,4713	2,3016	2,7388	2,3070	2,2165	2,3123	1,5088	2,3175	2,3224
	0,40	0,60	4,4238	4,2519	3,7786	4,2515	2,4600	4,2498	1,5346	4,2470	4,2429
	0,40	0,80	6,2865	6,7906	4,4057	6,7264	2,5656	6,6415	1,5445	6,5373	6,4158
	0,40	1,60	25,0000	9,5290	6,2500	8,4582	2,7778	7,1240	1,5625	6,5373	4,7343
	0,60	1,20	21,0784	12,4838	6,0907	12,0454	2,7635	11,4732	1,5614	10,8080	10,0895
	0,80	1,20	25,0000	27,8615	6,2500	27,5249	2,7778	26,9817	1,5625	26,2562	25,3788
	1,00	1,50	11,5591	17,8748	5,3716	17,9690	2,6909	17,9476	1,5554	17,8114	17,5655
	1,20	1,80	4,4238	6,7596	3,7786	6,8106	2,4600	6,8436	1,5346	6,8582	6,8540
	1,20	4,80	0,3315	0,9889	0,5451	1,2035	0,9056	1,4546	1,2106	1,7216	1,9567
	0,20	0,40	0,5346	0,4753	0,8406	0,4754	1,2231	0,4755	1,3256	0,4756	0,4757
8	0,40	0,60	1,0268	1,0097	1,4592	1,0131	1,6850	1,0163	1,4319	1,0193	1,0222
	0,40	0,80	1,5682	2,1153	2,0154	2,1817	1,9630	2,2422	1,4763	2,2955	2,3403
	0,40	1,60	25,0000	5,2429	6,2500	4,1241	2,7778	3,0421	1,5625	2,2249	1,6538
	0,60	1,00	12,9273	4,9281	5,5245	4,5554	2,7075	4,0913	1,5568	3,6021	3,1336
	0,80	1,20	25,0000	7,1715	6,2500	6,9824	2,7778	6,6885	1,5626	6,3163	5,8946
	1,00	1,50	3,6567	3,9928	3,4326	4,0685	2,3902	4,0959	1,5277	4,0730	4,0014
	1,20	1,80	1,0268	1,6419	1,4592	1,6789	1,6850	1,7093	1,4319	1,7321	1,7468
	1,20	4,80	0,0668	0,2370	0,1168	0,3310	0,2442	0,4850	0,6354	0,7450	1,1607
	0,20	0,40	0,2815	0,2480	0,4683	0,2480	0,8078	0,2480	1,1635	0,2480	0,2481
	0,40	0,60	0,5459	0,5477	0,8563	0,5489	1,2373	0,5501	1,3298	0,5512	0,5523
12	0,40	0,80	0,8427	1,3113	1,2422	1,3807	1,5464	1,4504	1,4051	1,5196	1,5870
	0,40	1,60	25,0000	5,1614	6,2500	3,7195	2,7778	2,5379	1,5625	1,7566	1,2588
	0,60	1,20	8,9549	3,9452	4,9922	3,5925	2,6461	3,1303	1,5516	2,6541	2,2207
	0,80	1,20	25,0000	4,6591	6,2500	4,4703	2,7778	4,1876	1,5625	3,8469	3,4826
	1,00	1,50	2,0494	2,4670	2,4274	2,5702	2,1187	2,6319	1,4970	2,6457	2,6100
	1,20	1,80	0,5459	1,0266	0,8563	1,0671	1,2377	1,1038	1,3298	1,1356	1,1617
	1,20	4,80	0,0348	0,1429	0,0614	0,2080	0,1329	0,3253	0,4112	0,5563	1,0220

TABLE 2. Relative efficiency of M_w with respect to $\hat{\beta}$.

r	$\frac{\beta_1}{\beta}$	$\frac{\beta_2}{\beta}$	W = 0,20				W = 0,80			
			K=0,20	K=0,40	K=0,60	K=0,80	K=0,20	K=0,40	K=0,60	K=0,80
4	0,20	0,40	2,5904	3,3579	4,0385	2,8588	1,7915	2,3309	3,1737	2,9370
	0,40	0,60	5,3651	6,0543	5,0835	2,6543	3,3933	4,2640	4,5332	2,7808
	0,40	0,80	11,4214	9,2250	5,3903	2,4898	3,9215	4,7993	4,7450	2,7391
	0,40	1,60	10,7817	7,5831	3,8794	1,9843	7,6362	7,5539	5,2984	2,5706
	0,60	1,20	48,3892	12,0476	4,7526	2,1759	11,4214	9,2250	5,3903	2,4898
	0,80	1,20	40,3044	11,5137	4,6293	2,1460	29,8928	12,1407	5,2637	2,3400
	1,00	1,50	9,0170	7,0089	3,7611	1,9601	44,5022	11,7944	4,6914	2,1608
	1,20	1,80	3,4656	4,1522	3,0322	1,8083	13,0732	8,1997	4,0007	2,0093
	1,20	4,80	0,2130	0,4787	0,8983	1,1509	1,9727	2,8378	2,5477	1,6991
8	0,20	0,40	0,5379	0,7605	1,1910	1,6381	0,3654	0,4988	0,7977	1,3937
	0,40	0,60	1,1790	1,6335	2,0873	1,8207	0,7167	1,0196	1,5159	1,7405
	0,40	0,80	2,8178	3,2658	2,8606	1,8285	0,8373	1,1875	1,6969	1,7761
	0,40	1,60	2,1746	2,6116	2,1394	1,4738	1,7534	2,2919	2,4909	1,8358
	0,60	1,20	19,3888	6,7100	3,0322	1,6528	2,8178	3,2658	2,8606	1,8285
	0,80	1,20	12,8270	6,0172	2,9148	1,6273	10,6826	6,2574	3,2803	1,7691
	1,00	1,50	1,7890	2,2928	2,0223	1,4491	15,7781	6,3746	2,9750	1,6401
	1,20	1,80	0,6598	1,0802	1,3823	1,2887	2,6984	2,9917	2,2625	1,4989
	1,20	4,80	0,0405	0,0967	0,2491	0,6300	0,3733	0,6723	1,0375	1,1709
12	0,20	0,40	0,2829	0,4084	0,6868	1,2331	0,1914	0,2642	0,4388	0,9404
	0,40	0,60	0,6287	0,9222	1,3791	1,5788	0,3785	0,5555	0,9139	1,3968
	0,40	0,80	1,5501	2,0487	2,2257	1,6946	0,4433	0,6531	1,0510	1,4665
	0,40	1,60	1,1391	1,6238	1,7103	1,3766	0,9455	1,3476	1,7789	1,6542
	0,60	1,20	13,1676	5,7801	2,7962	1,5853	1,5501	2,0487	2,2257	1,6946
	0,80	1,20	7,7647	4,8809	2,6481	1,5574	6,7652	4,9949	2,9538	1,6903
	1,00	1,50	0,9355	1,3939	1,5833	1,3465	10,0319	5,3328	2,7245	1,5716
	1,20	1,80	0,3410	0,6045	0,9643	1,1518	1,4214	1,9114	1,8484	1,4071
	1,20	4,80	0,0209	0,0506	0,1405	0,4566	0,1926	0,3661	0,6799	1,0126